# Ali Oku

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## PROFESSIONAL SUMMARY

Computational Biologist and Machine Learning (ML) Research Scientist with 5+ years of experience and dual M.Sc. degrees in Computer Science and Bioinformatics, specializing in single-cell foundation models and transformer-based representation learning for genomics and multi-omics. Constructs biological training and evaluation datasets from heterogeneous consortium and public data, fine-tunes and benchmarks foundation models on transcriptomic data, designs capability evaluations for Large Language Model (LLM)-based biology models, and engineers large-scale computational biology pipelines on High-Performance Computing (HPC) and cloud. Co-developed an open-source deconvolution method at NYGC benchmarked across 8 datasets, led transcriptomic analysis of 2,725 tissue samples from 718 donors for a 41-site ALS consortium, and partnered with researchers and experimental biologists across 6 institutions and 2 pharmaceutical companies on biomarker, target, and patient-stratification work.

## TECHNICAL SKILLS

**Programming and Scientific Computing:** Python, PyTorch, R, SQL, Bash, Linux, Git, pandas, NumPy, scikit-learn, tidyverse, Java
**Machine Learning and Deep Learning:** foundation models, Transformers, Hugging Face Transformers, deep learning, multimodal representation learning, Variational Autoencoders (VAEs), Maximum Mean Discrepancy (MMD), transfer learning, fine-tuning, Principal Component Analysis (PCA), benchmark design, model validation, data leakage prevention (donor-level held-out splits, nested cross-validation), explainability (SHapley Additive exPlanations (SHAP), attention-weight extraction), XGBoost, random forests, Support Vector Machines (SVM), scvi-tools
**LLMs, Agents and AI Evaluation:** Large Language Models (LLMs) for biology (Cell2Sentence on Pythia and Gemma), capability evaluations of foundation models, eval dataset design, model failure-case analysis (zero-shot versus fine-tuned), automated model-tracking pipelines over OpenAlex, GitHub, and Hugging Face Hub APIs
**Computational Biology and Bioinformatics Methods:** computational biology, bioinformatics, genomics, transcriptomics, multi-omics, cell-type deconvolution (MuSiC, Dampened Weighted Least Squares (DWLS), BayesPrism), differential expression, batch correction (Harmony, ComBat-seq), variance partitioning, Weighted Gene Co-expression Network Analysis (WGCNA), analysis of high-throughput functional screens (CRISPR), literature mining and knowledge-base construction, scanpy, Seurat, DESeq2, limma, edgeR, QIIME 2
**Biological and Molecular Data Sources:** single-cell RNA sequencing (scRNA-seq), single-nucleus RNA sequencing (snRNA-seq), bulk RNA sequencing (RNA-seq), long-read RNA-seq, Cellular Indexing of Transcriptomes and Epitopes by Sequencing (CITE-seq), Assay for Transposase-Accessible Chromatin using sequencing (ATAC-seq), spatial transcriptomics, Whole-Genome Sequencing (WGS), Whole-Exome Sequencing (WES), DNA methylation, Copy Number Variation (CNV), 16S rRNA amplicon sequencing, Fluorescence-Activated Cell Sorting (FACS) ground-truth cell proportions, Human Endometrial Cell Atlas (HECA), PubMed, PubTator 3, Open Targets, g:Profiler, Gene Ontology (GO), Kyoto Encyclopedia of Genes and Genomes (KEGG), Reactome, The Cancer Genome Atlas (TCGA)
**Workflow, Cloud and Infrastructure:** pipeline development, workflow orchestration (Nextflow, Snakemake, Workflow Description Language (WDL)), HPC with Slurm, cloud computing (Amazon Web Services (AWS), Google Cloud Platform (GCP)), Docker, GitHub Actions, PyPI packaging, reproducible and version-controlled code, documentation
**Domain Expertise:** cancer genomics and tumor microenvironment, neurodegeneration (Amyotrophic Lateral Sclerosis (ALS)), autoimmune disease (systemic lupus erythematosus (SLE)), microbiology and microbiome science, translational biomarker discovery, genetic ancestry and cancer health disparities
**Experimental Biology and Design:** Next-Generation Sequencing (NGS) library preparation, microbial DNA extraction, Polymerase Chain Reaction (PCR), mammalian cell culture (Chinese Hamster Ovary (CHO), Human Embryonic Kidney (HEK), NS0), High-Performance Liquid Chromatography (HPLC), Gas Chromatography (GC), Fourier-Transform Infrared spectroscopy (FT-IR) under FDA Good Manufacturing Practice (GMP), experimental design with bench scientists (CRISPR knockdown, drug-treatment studies)

## PROFESSIONAL EXPERIENCE

**Senior Bioinformatics Analyst, Computational Biology | New York Genome Center (NYGC) | New York, NY | 11/2026 – Present**

**Bioinformatics Analyst, Computational Biology | New York Genome Center (NYGC) | New York, NY | 08/2023 – 10/2026**

- Co-developed DECONVersation at NYGC with fellow researchers as first author, through preprint and PyPI release, framing deconvolution as a capability benchmark for foundation models; 240+ downloads in 30 days.
- Validated 6 embedding models against FACS-measured cell proportions on 6 real bulk RNA-seq datasets and 2 pseudobulk sets, matching or beating MuSiC, DWLS, and BayesPrism on correlation and Root Mean Square Error (RMSE).
- Constructed ground-truth training and evaluation data where labeled bulk data is sparse by simulating 1,000 pseudobulk samples of 800 cells per donor, using donor-level held-out splits to prevent data leakage.
- Fine-tuned Geneformer and the LLM-based Cell2Sentence for cell-type classification, improving deconvolution over zero-shot embeddings in every benchmark case and documenting zero-shot failure cases that guided model selection.
- Benchmarked Geneformer, scGPT, and Cell2Sentence against scVI, Harmony, and PCA for batch correction and representation learning across 10 datasets and 2M+ cells, setting the group's model-selection criteria.
- Led transcriptomic analysis for the NYGC ALS Consortium (2,725 postmortem tissue samples, 718 donors, 15 central nervous system regions), linking expression and cell-type composition to 5 clinical and genetic variables.

- Performed QC, differential expression, and deconvolution for a Mount Sinai long-read RNA-seq study of TDP-43 knockdown in induced pluripotent stem cell (iPSC) neurons that identified 20,355 novel isoforms.
- Deployed the public ALS Gene Browser (R Shiny), giving about 500 researchers across the USA, Europe, and Japan no-code access to consortium expression results.
- Quantified genetic-ancestry effects on tumor expression in 142 breast cancer patients (SAMBAI) by integrating WGS and RNA-seq with variance partitioning; first author, American Association for Cancer Research (AACR) 2026.
- Performed QC, differential expression, deconvolution, and survival association analyses for Polyethnic-1000 (851 tumor-normal genomes, 8 cancer types) and a metastatic prostate cancer phylogeny study.
- Engineered Nextflow and WDL pipelines for 4 assay types plus CRISPR perturbation-screen analysis on Slurm HPC and cloud, returning results fast enough to inform the next experiment.
- Established the group's single-cell quality control (QC) framework with automated metrics, doublet and ambient-RNA handling, and versioned reference data, making results reproducible across teams.
- Communicated results to biologists and ML researchers through 4 first-author conference presentations and interactive R Shiny and D3.js dashboards used across 6 partner institutions.

**Bioinformatics Intern, Machine Learning | EMD Serono (Merck KGaA) | Billerica, MA | 02/2023 – 07/2023**
- Curated and integrated 4 omics modalities for 5,000+ tumor samples across 20 cancer types from The Cancer Genome Atlas (TCGA) into a training set for tumor microenvironment (TME) classification.
- Engineered a cancer-agnostic multimodal ML model classifying TME subtypes and applied it to an internal colorectal cancer cohort of about 100 patients, transferring from public to proprietary data.
- Validated translational relevance with Kaplan-Meier analysis linking predicted TME subtypes to 2 clinical endpoints, patient survival and immunotherapy response, supporting biomarker discovery.
- Benchmarked 3 model classes (XGBoost, random forest, neural network) with nested cross-validation and SHAP explainability, delivering the evaluation to the research team.

**Bioinformatics Intern, Computational Oncology | Takeda Pharmaceuticals | Cambridge, MA | 05/2022 – 08/2022**
- Quantified drug-treatment effects at single-cell resolution by comparing treated and vehicle cells across 2 single-cell modalities (scRNA-seq and CITE-seq), working with the bench team that ran the study.
- Discovered colorectal cancer subtype-associated gene modules using WGCNA on bulk RNA-seq from 150 patients, with pathway and GO enrichment.
- Built logistic regression subtype classifiers from expression and TME features, reaching 85% accuracy on held-out samples.

**Machine Learning Intern | Max Delbrück Center for Molecular Medicine | Berlin, Germany | 05/2021 – 08/2021**
- Designed a multimodal variational autoencoder integrating 3 omics modalities with an MMD alignment loss to improve cross-modal integration of colorectal cancer data.
- Validated the learned embeddings as predictors of 2 clinical outcomes, survival and drug response, across colorectal cancer subtypes.

**Graduate Research Assistant, Microbiome and Computational Biology | Northern Illinois University | DeKalb, IL | 08/2020 – 05/2023**
- Built an end-to-end Python and R pipeline for about 500 16S rRNA microbiome samples, applying ML and network analysis to link microbial composition to prairie restoration practices (M.Sc. thesis).
- Supervised about 5 undergraduate and graduate researchers through sample collection, QIIME 2 processing, and analysis.

**Genomics Lab Technologist | Diversigen (OraSure Technologies) | 2020**
- Performed high-throughput microbial DNA extraction, PCR, and NGS library preparation for clinical microbiome studies and authored extraction standard operating procedures (SOPs).

**Cell Culture Research Associate | Bio-Techne | 2019 – 2020**
- Optimized growth conditions for 3 mammalian cell lines (NS0, CHO, HEK) with purification and molecular biology teams to resolve protein expression issues.

**Quality Control Chemist | Perrigo | 2018 – 2019**
- Ran HPLC, GC, ultraviolet (UV), and FT-IR analyses of pharmaceutical products under FDA GMP, leading internal audits and authoring SOPs and deviation reports.

## EDUCATION

**M.Sc., Biological Sciences and Bioinformatics | Northern Illinois University | DeKalb, IL | 2023**
- Thesis: The Role of Machine Learning and Network Analyses in Understanding Microbial Composition in an Experimental Prairie

**M.Sc., Computer Science | Northern Illinois University | DeKalb, IL | 2022**

**B.Sc., Biological and Biomedical Sciences | Minnesota State University, Mankato | Mankato, MN | 2017**

## PUBLICATIONS AND OPEN-SOURCE


- Oku A, Geiger H, Robine N, Fu R. DECONVersation: Single cell foundation model-derived embeddings for robust cell type deconvolution of bulk RNA-seq. bioRxiv (2026). doi:10.64898/2026.10.02.756356
- Humphrey J, Oku A, Byrska-Bishop M, et al., Phatnani H. The New York Genome Center ALS Consortium resource integrates postmortem tissue transcriptomics and whole genome sequencing to empower biological discovery. medRxiv (2026). doi:10.64898/2026.04.29.26350889
- Nauseef JT, Chu TR, Hooper WF, Alonso A, Oku A, et al. A complex phylogeny of lineage plasticity in metastatic castration-resistant prostate cancer. npj Precision Oncology 9, 91 (2025). doi:10.1038/s41698-025-00854-4

- Frimer M, Gee D, Goldstein ZR, et al. Comprehensive molecular characterization of high-grade endometrial cancer in an ancestrally diverse cohort. bioRxiv (2026). doi:10.64898/2026.05.01.721962
- Subhash S, Bammert MT, Yueh B, et al. A high-fidelity and ancestrally inclusive patient-derived organoid platform resolves cancer cell plasticity in uterine carcinosarcoma. bioRxiv (2026). doi:10.64898/2026.04.21.720027
- Robine N, Winterkorn L, Chu T, Goldstein Z, Hooper W, Oku A, et al. Polyethnic-1000: Advancing cancer genomics by studying ethnically diverse, underserved patient populations in New York. Cancer Epidemiology, Biomarkers and Prevention 33 (2024). doi:10.1158/1538-7755.DISP24-C102
- Oku A, et al. First-author presentations: Machine Learning in Computational Biology (MLCB) 2025; Mid-Atlantic Bioinformatics Conference 2025; RNA Symposium 2025; AI in Biology Conference 2026; AACR Science of Cancer Health Disparities 2026.
- Released dataset: NYGC ALS Consortium transcriptomic and genomic data, Zenodo (zenodo.org/communities/nygc_als), co-author.
- Tool: DECONVersation, open-source Python package for deconvolution with Geneformer, Cell2Sentence, scGPT, and CellHermes embeddings (github.com/nygenome/DECONVersation, PyPI).
- Tool: scFM Tracker, automated weekly evaluation of 19 single-cell foundation models on citations, usage, openness, and runnability (eastmanmd.github.io/scFM-tracker).
- Tool: Lupus Gene Explorer, literature-mined knowledge base ranking 1,505 SLE genes from about 114,000 PubMed articles using PubTator 3 entity normalization and Open Targets evidence (eastmanmd.github.io/LupusExplorer).
- Tool: ALS Gene Browser, public R Shiny app for NYGC ALS Consortium transcriptomics (alsbrowser.nygenome.org).
- Tool (in development): TMEmbed, Python package for TME subtype classification on scVI embeddings of multi-cohort bulk expression, currently tested on synthetic data (github.com/Eastmanmd/TMEmbed).